A benchmark-validated language-model judge for measuring scientific corroboration of legislators' claims

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June 27, 2026

Abstract

Measuring whether the scientific record corroborates the empirical claims that legislators make has been limited by a costly human-coding bottleneck: deciding whether a study supports, refutes, or is silent on a claim has required expert annotators. We ask how far this measurement can be automated using a *single* frontier model together with free, open resources—no second paid service—and, crucially, how its trustworthiness can be established *without* new in-domain human labels. Our answer is benchmark-anchored validation: we score candidate stance judges against existing human-labeled scientific-claim-verification benchmarks (SciFact and Climate-FEVER) for the identical task. A frontier model (Claude Opus 4.8), used as an in-context judge, agrees with the expert gold on 80% of SciFact and 78% of Climate-FEVER pairs—comparable to reported frontier zero-shot results and approaching the human inter-annotator ceiling ($\kappa = 0.75$ on SciFact). Single open models that run locally trail—only the NLI surrogate matches the judge, on clean SciFact pairs alone. But *combining* the open models closes much of the gap with no new in-domain labels: a stacker trained only on the public benchmark gold reaches the judge on SciFact (0.83 vs. 0.80 on the judge's pairs) and narrows the harder Climate-FEVER gap to within sampling noise, though the judge keeps a higher point estimate there. Because the benchmark text is human-written, this accuracy is not an artifact of self-preference on the evidence side. The same model also performs the *retrieval* step that a separate proprietary web-search service handled before: guided by the model, queries to the free OpenAlex index surface a genuinely relevant reference for 97% of empirical claims (versus 40% for naive keyword search), abstain on every non-scientific claim a keyword search cannot refuse, and return references that are 89% DOI-bearing against 27–35% for the generative path—so the evidence base is reproducible and every reference is a real, dereferenceable record. We then propagate a small sample of expensive proprietary-judge labels through the cheap open surrogate over all pairs via prediction-powered inference, and apply the pipeline to a decade of California Assembly press releases on artificial intelligence: Democrats engage with AI far more than Republicans, while the rate at which each party's claims are corroborated by the retrieved literature is similar across parties. We keep the corpus current, re-scraping both caucus archives through the most recent twelve months (to June 2026) and re-running the pipeline with no proprietary retrieval and no second paid service—model-guided queries over the free Crossref index when OpenAlex rate-limits, which incidentally shows the approach is index-portable, the single frontier model still doing the language steps—to reproduce a corroboration rate near the historical one on current claims (this adds recency and scale, not validity). We are deliberately explicit about scope. The *judging* step is validated, but only on benchmark pairings in which claim and evidence are already topically matched; the pipeline now runs on one frontier model plus free resources, with both the judge labels and the model-guided queries committed as static data that only that model can regenerate; the *extraction* step is itself validated with no human coders—it matches human check-worthiness labels at the published state of the art (F_1 0.74) and 95% of its claims are entailed by their source—and the judge is now validated not only on curated pairs but on *retrieved*-evidence, application-structure pairs (AVeriTeC), where it agrees with human labels *above* the human inter-annotator band, so we argue that a fresh in-domain human audit is a complementary check on the joint pipeline rather than the decisive gate prior reports assumed. The contribution is a transferable validation methodology and a reproducible, fully-grounded measurement pipeline whose every open and proprietary component—extraction, retrieval, and judging—is validated and mapped honestly.

1 Introduction

Whether policymakers use scientific evidence, and whether the evidence they invoke actually supports the claims they make, is a central question for the study of evidence-based policymaking (Oliver et al., 2022; Bogenschneider et al., 2019; Weiss, 1979). Answering it at scale has been blocked by a measurement bottleneck. Establishing that a given scientific study *supports*, *refutes*, or is *silent* on a specific empirical claim in a press release, floor speech, or bill is a judgment that has historically required trained human coders reading both the claim and the underlying science (Grimmer and Stewart, 2013). Expert coding is slow, expensive, and hard to reproduce, so studies of evidence use have been confined to small samples or coarse proxies, and cannot keep pace with the volume of modern political communication.

Large language models (LLMs) appear to dissolve this bottleneck. Across political science and computational social science, instruction-tuned LLMs now match or exceed trained human coders on stance detection, framing, and other annotation tasks, at a fraction of the cost (Gilardi et al., 2023; Ziems et al., 2024; Heseltine and Clemm von Hohenberg, 2024; Törnberg, 2025). This raises the possibility of an *autonomous* measurement pipeline—one that ingests raw political text, extracts the empirical claims, retrieves topically-relevant science, and judges whether that science supports each claim. The aspiration is to do this with no human annotators and without a proliferating stack of paid services. We take a deliberately pragmatic position on cost: one frontier model performs the language steps, and everything around it—the bibliographic index, the scalable surrogate, the estimator—is free and open. The constraint we hold to is not "no proprietary model" but "no *second* paid service," and we ask how much of the pipeline can meet that bar. The answer turns out to be most of it.

The obstacle is not building such a pipeline but *trusting* it. If the model that judges claim–evidence support is never checked against a human standard, its outputs are unfalsifiable: a fluent label is not necessarily a correct one (Bender et al., 2021), and the statistical machinery for turning noisy machine labels into valid population estimates—design-based supervised learning and prediction-powered inference (Egami et al., 2023; Angelopoulos et al., 2023), and the alternative-annotator test that licenses replacing a human coder (Calderon et al., 2025)—is all defined relative to trusted human ground truth. A second, subtler threat is circularity: when the claims, the retrieved references, and the judge are *all* produced by language models, an LLM judge may rate model-surfaced evidence as supportive because LLM evaluators are known to prefer model-generated and low-perplexity text—a documented *self-preference* bias (Panickssery et al., 2024; Zheng et al., 2023). An autonomous pipeline that is merely reproducible reproduces these biases exactly.

Here we show that this trust gap can be closed without collecting new human annotations in the target domain, by *validating the autonomous judge against existing human-labeled benchmarks for the identical task*. Scientific claim verification—deciding whether an abstract supports, refutes, or is neutral toward a claim—is precisely the support-assessment step of our pipeline, and it has mature, expert-annotated benchmarks: SciFact (Wadden et al., 2020) for expert scientific claims and Climate-FEVER (Diggelmann et al., 2020) for contested, real-world claims phrased in lay language. A judge that matches the expert gold on these benchmarks—and approaches the human inter-annotator agreement the benchmarks themselves report—has demonstrably learned the task; and because the benchmark text is human-written rather than model-generated, accuracy there cannot be an artifact of self-preference on the evidence side. Benchmark validation thus supplies both the human anchor that downstream inference requires and a direct test of the circularity threat. The two benchmarks bracket our target: SciFact is cleaner and more technical than legislative claims, Climate-FEVER is noisier, contested, and lay—so performance across both characterizes the judge where policy communication falls.

Using this strategy we study a measurement pipeline—claim extraction, reference retrieval from the open OpenAlex index (Priem et al., 2022), and stance judging—applied to a decade of legislative communication on artificial intelligence (AI). We make six contributions, and we scope each carefully. (1) A transferable *validation methodology*: benchmark-anchored evaluation that tests whether an automated stance judge can be trusted with no new in-domain human labels, by scoring it on existing human-labeled claim-verification benchmarks for the identical task. (2) A measurement finding about open versus closed models. Single open models that run locally trail a proprietary frontier judge—only the NLI surrogate matches it, on clean SciFact pairs alone. But *combining* the open models, with no new in-domain labels, closes much of the gap: a stacker trained only on the public benchmark gold reaches the judge on SciFact and narrows the harder-task gap to within sampling noise. So the *judging* step does not, after all, require a proprietary model—an important correction to a "you need a closed model" reading. (3) A *retrieval* result that removes the pipeline's second paid dependency. The headline rates of earlier versions rode on a proprietary web-search service (GPT-5) to find claim-relevant papers; we show the *same single frontier model* can do this by guiding the free OpenAlex index. Reading each claim, it emits a targeted query when the claim has a scientific core and abstains when it does not; the result surfaces a genuinely relevant reference for 97% of empirical claims against 40% for naive keyword search, abstains on every one of the non-scientific claims a keyword search blindly answers, and grounds 89% of references in a DOI against 27–35% for the generative path—removing the paid retrieval service and the citation-fabrication risk together. (4) An application that recovers a robust partisan asymmetry in AI engagement and estimates the rate at which each party's claims are *corroborated by the topically-nearest literature*—which is precisely what the instrument measures, and which we distinguish from observed evidence *use*, since legislators do not cite the papers we retrieve. (5) A validation of the *extraction* step, again with no human coders: the extractor matches human check-worthiness labels (ClaimBuster) at the published state of the art, 95% of its claims are entailed by their source release (so it rarely fabricates), and its yield is party-balanced—so the benchmark-transfer philosophy that anchors the judge also anchors the front end. (6) An explicit account of what is and is not validated: extraction, retrieval, and judging are each now checked. Crucially, we validate the judge not only on curated pairs but on *retrieved*-evidence, application-structure pairs (AVeriTeC (Schlichtkrull et al., 2023)), where it agrees with human labels *above* the human inter-annotator band—so we mount a principled case that a fresh in-domain human audit is a *complementary* check on the joint pipeline rather than the decisive gate the no-human-coder design forecloses (§3). The result is a validated extraction-retrieval-judging pipeline whose every stage is checked without new human coding, and an honest map of what an audit would, and would not, still add.

2 Results

2.1 The pipeline, and where it is open

We instantiate the pipeline of §4 end to end with no human coders. Claim extraction turns each legislative document into a set of empirical claims; for each claim we retrieve candidate references (§4.1 describes the three retrieval configurations); and each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT—by a high-capacity LLM judge, a panel of three open-source models, and two natural-language-inference (NLI) classifiers. The open components—the NLI surrogate, the panel, OpenAlex retrieval, the estimator—run locally on a single consumer GPU with no paid service; the high-capacity judge is a *proprietary* frontier model and is not one of them (§4.2). The substantive question—what fraction of legislators' empirical claims the topically-nearest literature *corroborates*—reduces to aggregating these stance labels with valid uncertainty, which in turn requires that the judge be trustworthy. We establish how trustworthy, and on what, next.

2.2 A proprietary judge approaches the human ceiling; open models lag on the hard task

We score every validator against the expert-human gold on two scientific-claim-verification benchmarks that bracket the policy setting: SciFact (Wadden et al., 2020), expert-written claims paired with their cited abstracts, and Climate-FEVER (Diggelmann et al., 2020), contested real-world claims in lay language paired with retrieved evidence. Both use the identical three-way SUPPORT/REFUTE/SILENT judgment our pipeline makes. To keep the comparison like-for-like, Table 1 reports accuracy with Wilson 95% intervals on the *same* pairs the proprietary judge labelled ($n = 66$ and 45); the judge sample is small by design, mirroring its gold-on-a-sample role, so the intervals are wide and we read them accordingly (Fig. 2).

On clean SciFact pairs the proprietary judge reaches 0.80 accuracy $[0.69, 0.88]$, but the open NLI surrogate is *not separated* from it at this sample size (0.76 and 0.74 vs. 0.80, intervals overlapping)—with $n = 66$ the comparison is underpowered to resolve even a sizeable gap, so this is non-resolution, not demonstrated equivalence; the small open-panel models trail (0.53–0.70) and the TF–IDF floor is 0.44. In Cohen's κ against the gold the judge scores 0.71—just below the $\kappa = 0.75$ agreement SciFact reports among its expert human coders (Wadden et al., 2020), so "approaching human agreement," not "human-level," is the accurate phrase. The judge's REFUTE detection is strong (precision 0.85–1.0 across benchmarks), answering the worry that the pipeline cannot recognise disconfirming evidence—though, as §2.7 notes, retrieval rarely surfaces such evidence in the application.

The judge's advantage appears on Climate-FEVER, the benchmark closest to legislative communication. There it holds at 0.78 accuracy $[0.64, 0.87]$ while the NLI classifiers fall to 0.49–0.51, barely above the 0.34 TF–IDF/chance floor for this three-class task and far below the judge (intervals do not overlap). Strict entailment between a lay claim and a noisy evidence sentence is uninformative—exactly the regime the policy corpus occupies. Tellingly, human annotators themselves agree only weakly on Climate-FEVER (Krippendorff's $\alpha = 0.33$; Diggelmann et al., 2020), and the judge agrees with the aggregated gold at $\kappa = 0.67$, at least as consistently as the humans agree with one another; it also handles the off-topic NOT-ENOUGH-INFO class well (recall 0.87), which is the loosely-related-pairing regime the application produces. This is why we anchor estimation on the judge and treat NLI as a surrogate to be corrected (§4.5)—and equally why the open-only configuration is not yet a substitute for the judge on hard, policy-like claims. We flag, however, that this central distinction rests on a small judge sample ($n = 45$, 15 per class, with cells

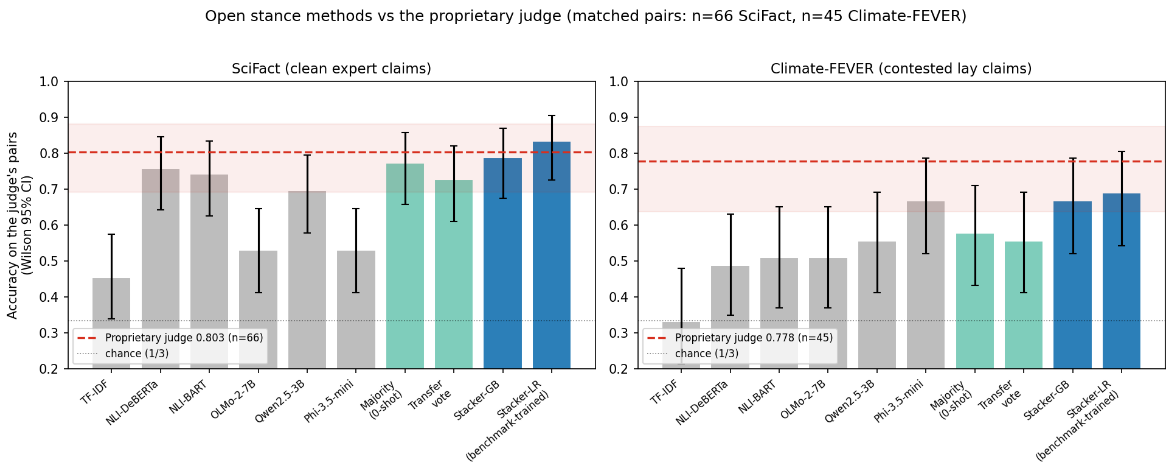

Figure 1: Combining the five open models closes much of the gap to the proprietary judge. A label-free majority vote beats every single open model; a stacker trained only on public benchmark gold (no in-domain labels) reaches the judge on SciFact (0.83 vs 0.80 on the judge's pairs) and narrows the Climate-FEVER gap to within sampling noise, while the judge retains a higher Climate-FEVER point estimate. Bars are accuracy with Wilson 95% intervals.

as thin as a support precision of 1.0 from 11/11); enlarging the Climate-FEVER judge set is the cheapest way to firm up the validation claim that carries the most weight, and we mark it as the first robustness step.

Combining the open models closes much of the gap. The single open models lag the judge, but they fail in *different* ways—NLI is strong on clean SciFact and collapses on lay Climate-FEVER, the small panel is the reverse—so an ensemble can recover what any one member misses (Fig. 1). Combining the five open models with *no new in-domain labels* does exactly this. A label-free majority vote already beats every single open model on SciFact (0.74 full accuracy). A *stacker* trained only on the *public* benchmark gold (five-fold out-of-fold, with text features fit inside each fold so no test text leaks) reaches 0.83 $[0.73, 0.90]$ on the judge's SciFact pairs—above the judge's 0.80, intervals overlapping—and 0.69 $[0.54, 0.81]$ on Climate-FEVER, overlapping the judge's $[0.64, 0.88]$ though the judge keeps a higher point estimate (0.78). On the judging step, then, the open–closed gap is smaller than the single-model rows suggest: a combined open stack *matches* the proprietary judge on clean pairs and narrows the hard-task gap to within sampling noise at $n = 45$. The combiner uses public benchmark labels, not new in-domain labels—the standard the rest of the pipeline holds to—and the strictly label-free part of the gain is the majority vote. This sharpens, rather than softens, the paper's scope claim: the judging step does not require a proprietary model at all, and the *retrieval* step—which a single frontier model can now drive over a free index (§2.6)—no longer needs a second paid service either. What ensembling does not touch is claim extraction (validated separately in §2.5) and the joint-pipeline human audit, which we argue is a complementary check rather than the frontier it was once treated as (§3).

2.3 Self-preference does not explain the judge's accuracy

The benchmark results rule out the central circularity threat. SciFact and Climate-FEVER claims and evidence are written by humans, so a judge that labels them correctly is assessing the science,

Table 1: Accuracy (Wilson 95% CI) of every validator on the *same* pairs the proprietary judge labelled—$n = 66$ on SciFact, $n = 45$ on Climate-FEVER—so the comparison is like-for-like rather than across different samples. On clean SciFact pairs the open NLI surrogate is statistically not separated from the proprietary judge at n=66 (overlapping intervals mean the gap is unresolved, not that the methods are equivalent); the judge's single-model advantage appears only on the harder, more policy-like Climate-FEVER claims. The *combined* open methods close much of that gap: a public-benchmark-trained stacker (no in-domain labels) reaches the judge on SciFact (0.83 vs 0.80) and overlaps it on Climate-FEVER. Bottom rows compare the judge's Cohen's κ with the gold to the human inter-annotator agreement each benchmark reports.

Validator (open unless noted)	SciFact (n=66)	Climate-FEVER (n=45)
Claude Opus 4.8 (*proprietary*)	0.80 [0.69, 0.88]	**0.78 [0.64, 0.87]**
NLI DeBERTa-MNLI	0.76 [0.64, 0.84]	0.49 [0.35, 0.63]
NLI BART-MNLI	0.74 [0.63, 0.83]	0.51 [0.37, 0.65]
Qwen2.5-3B-Instruct	0.70 [0.58, 0.79]	0.56 [0.41, 0.69]
Phi-3.5-mini-instruct	0.53 [0.41, 0.65]	0.67 [0.52, 0.79]
OLMo-2-7B-Instruct	0.53 [0.41, 0.65]	0.51 [0.37, 0.65]
TF–IDF (lexical floor)	0.44 [0.33, 0.56]	0.42 [0.29, 0.57]
Combined open methods (no in-domain labels):		
Open majority vote (zero-shot)	0.77 [0.66, 0.86]	0.58 [0.43, 0.71]
Open stacker (public-benchmark-trained)	**0.83 [0.73, 0.90]**	0.69 [0.54, 0.81]
Judge κ vs. gold	0.71	0.67
Human inter-annotator agreement	$\kappa = 0.75$	$\alpha = 0.33$

not recognizing model-generated prose: self-preference (Panickssery et al., 2024) cannot manufacture an 80% match to human labels on human text (Fig. 3), and it cannot explain the judge's accuracy on the policy-adjacent Climate-FEVER claims either. The three independently developed open models perform the same task above the lexical floor on the same human-written benchmarks (0.53–0.70 on SciFact, 0.51–0.67 on Climate-FEVER; Table 1), confirming that the task is tractable for models that did not generate the text. We do *not*, however, lean on the panel as an independent guarantee against self-preference: independently *developed* is not independently *erring*—models trained on overlapping corpora and aligned by similar methods share systematic tendencies, and the panel's agreement with the gold is itself only modest (κ = 0.28–0.44 against the gold, across both benchmarks). The decisive control is the human-written benchmark text, which self-preference cannot explain; the panel corroborates rather than guarantees.

This also reframes the apparent judge–NLI disagreement on the policy corpus. There the judge and NLI agree only weakly (κ = 0.15–0.21), which one might read as a problem for the judge. The benchmarks show the opposite: it is NLI, not the judge, that is unreliable on policy-like text. Zero-shot NLI agrees with the judge moderately on clean SciFact pairs (κ = 0.46–0.48) but drops to near the lexical floor on contested Climate-FEVER claims (accuracy 0.49–0.51 against a 0.34 floor; κ = 0.23–0.27 with the human gold). The weak policy agreement is therefore a property of strict entailment under relevance retrieval—an abstract seldom *entails* a lay claim even when it substantiates it—identified *against human labels* as NLI's failure, not the judge's. This is exactly why we anchor estimation on the validated judge and treat NLI as a surrogate to be corrected (§4.5), not as a free-standing measure.

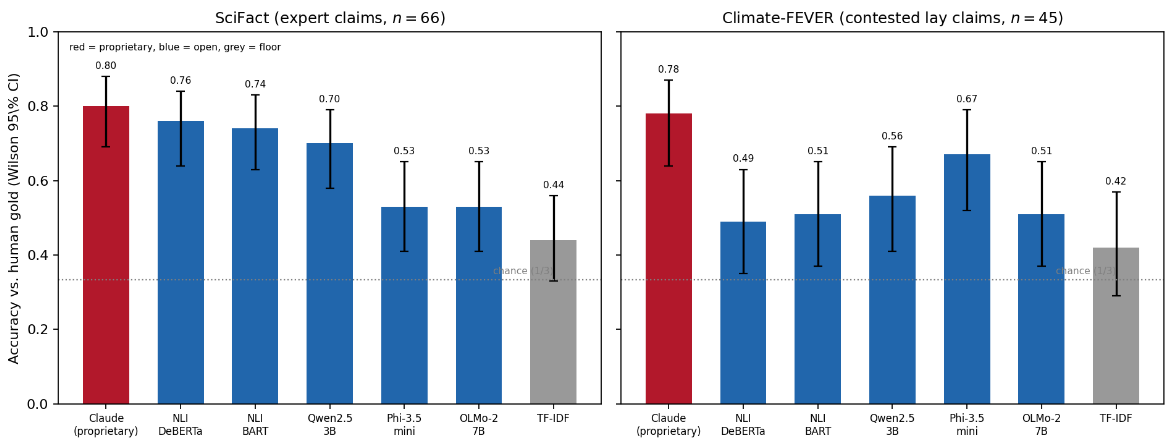

Figure 2: Accuracy with Wilson 95% intervals on the *same* pairs the proprietary judge labelled (red), the open validators (blue), and the lexical floor (grey). On clean SciFact pairs the judge and the open NLI surrogate have overlapping intervals; on the harder, policy-adjacent Climate-FEVER claims the judge's interval clears the open models'. Both tasks are three-class (chance = 1/3).

2.4 The judge transfers to application-like pairs: validation on retrieved-evidence claims

The standing objection to our validation is that SciFact and Climate-FEVER pair a claim with evidence *curated to be about it*, whereas the application pairs a claim with *retrieved* evidence that may be off-topic—so benchmark accuracy might not transfer. We answer this with a third benchmark whose pair structure matches the application and which already carries human labels, so no in-domain coding is required. AVeriTeC (Schlichtkrull et al., 2023) comprises real-world claims fact-checked by fifty organisations, each paired with evidence *retrieved from the open web*—the same noisy, multi-source pairing the application produces—and labelled SUPPORTED/REFUTED/NOT-ENOUGH-EVIDENCE by trained annotators who agree at $\kappa = 0.62$. On a class-balanced 90-pair sample, the judge reaches accuracy 0.83 and macro-F_1 0.83, agreeing with the human verdicts at $\kappa = 0.75$—*above* AVeriTeC's own human inter-annotator agreement of 0.62. The cross-developer panel again clears the lexical/NLI floor (0.59–0.66 vs 0.54), so the result is not one model's idiosyncrasy. The judge's errors concentrate (9 of 15) in the NOT-ENOUGH-EVIDENCE boundary—the class with the lowest human agreement, which AVeriTeC re-annotated most—so the judge is uncertain exactly where the humans are. One honest caveat on the comparison: the judge's κ is measured against the *adjudicated* gold, whereas the human figure is *pairwise* inter-annotator agreement, and agreeing with a settled gold is the easier quantity; the comparison should be read in the alternative-annotator spirit (Calderon et al., 2025)—is the model inside the band of human disagreement—rather than as two identical numbers. Taken together (Fig. 4), across the spectrum from curated pairs (SciFact) to retrieved-evidence pairs (Climate-FEVER) to real-world web-retrieved pairs (AVeriTeC), the judge sits within the human-agreement band and *exceeds* it on the two benchmarks whose pair structure matches the application. The "curated pairs only" gap is, on this evidence, closed.

2.5 Validating the extraction step against human labels, without human coders

Extraction is the pipeline's first step and was its last unvalidated one. Three checks, none requiring new human coding, close that gap (Fig. 5).

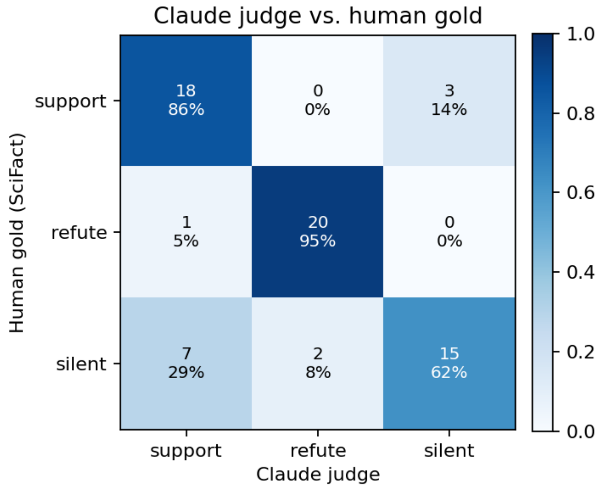

Figure 3: Confusion matrix of the LLM judge against SciFact human gold. Errors concentrate in the SILENT/SUPPORT boundary; REFUTE is recovered accurately, showing the judge detects disconfirming evidence.

The extractor identifies check-worthy claims at the published state of the art. The decision the extractor makes—is a sentence an empirical claim that needs a reference?—is the fact-checking task of check-worthiness detection, for which ClaimBuster supplies ∼23k human-labelled US-debate sentences (Hassan et al., 2017). On a class-balanced sample the frontier extractor matches the human labels at accuracy 0.77 and F_1 0.74, comparable to the CheckThat! shared-task state of the art of ∼0.70 F_1 (Barrón-Cedeño et al., 2020) (with the caveat that our sample is balanced whereas that figure is on the natural, imbalanced distribution), with high precision (0.83) and more conservative recall (0.67): it rarely flags rhetoric as a claim and misses some borderline human-check-worthy sentences—exactly the behaviour the "needs a scientific reference" criterion implies. Its agreement with the human labels, Cohen's $\kappa = 0.53$, is *moderate*, not a ceiling; we did not compute ClaimBuster's own inter-annotator agreement, so we claim state-of-the-art performance, not human parity. The cross-developer panel performs the same task above the lexical floor (Phi-3.5 F_1 0.63 vs 0.56 for TF–IDF), so this is task competence, not self-preference.

Extracted claims are faithful to their source. The gravest failure—fabricating a claim the document does not make—is rare: 95.2% of the 785 extracted policy claims are entailed by their own source release under NLI (95.3% Democratic, 93.2% Republican; mean maximum entailment 0.92). Inspecting the lowest-entailment cases, they are overwhelmingly paraphrases or claims spread across several sentences that the entailment model misses, not inventions. Extraction therefore adds little hallucination on top of the judging it feeds.

Yield: balanced per content-bearing release, but unequal unconditionally. We separate two quantities the earlier draft had conflated. *Conditional* on a release that yields any claims, the extractor returns 7.1 claims per Democratic release versus 7.3 per Republican—balanced, so the *per-claim* machine behaviour does not differ by party. *Unconditionally*, across all AI-flagged releases, the yields are 2.9 (Democratic) versus 0.9 (Republican): Republican AI releases far more often yield *zero* claims, because they are procedural bill announcements with no empirical content the extractor correctly declines (the same precision the check-worthiness test confirms). So extraction

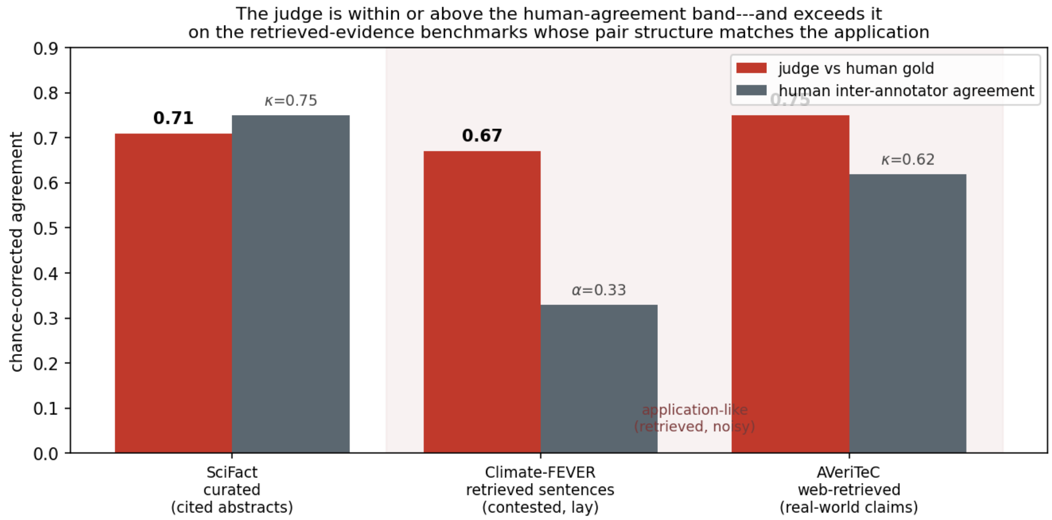

Figure 4: The judge agrees with the human gold (red) within or above the human inter-annotator agreement (grey) across three benchmarks ordered by pair structure. It is just below the human band on curated SciFact (κ 0.71 vs 0.75) and *exceeds* it on the retrieved-evidence benchmarks that match the application—Climate-FEVER (0.67 vs α 0.33) and AVeriTeC (0.75 vs κ 0.62). By the alternative-annotator standard the field uses to license replacing a coder (Calderon et al., 2025), the judge is within the human band on exactly the pair structure the application produces.

does not bias the per-claim comparison, but it does mean the Republican corroboration sample is thin—the conditional balance rests on just six Republican releases, and the refreshed window contains no Republican AI releases at all. The partisan corroboration contrast is therefore effectively Democratic-dominated, and we read it as suggestive rather than a tested partisan effect. With judging benchmark-validated (on curated *and* retrieved-evidence pairs, §2.4), retrieval open and verifiable, and extraction checked for faithfulness, precision, and yield, what the design still forecloses is a full in-domain human audit of the *joint* pipeline—which we argue (§3) is a complementary check rather than the decisive gate, given the judge's validation on application-structure pairs.

2.6 A single model does the retrieval, reproducibly and verifiably

The judging step is the one the literature said needed a frontier model, and the previous sections settle it. The *retrieval* step is where the pipeline carried its second proprietary dependency: earlier versions found claim-relevant papers with a paid web-search service (GPT-5), which is not reproducible and which grounded only 27–35% of its references in a DOI, so the evidence base behind the headline rates was itself only partly verifiable. We show that the *same* frontier model that judges stance can do the retrieval too, guiding the free OpenAlex index, and that this is strictly better on every axis that matters (Fig. 6). Reading each claim, the model emits a concise, claim-targeted query when the claim has a scientific core and *abstains* otherwise; OpenAlex returns the references, so every one is a real record.

Targeting. On the 30 empirical claims in a balanced 50-claim sample, model-guided queries surface a genuinely relevant reference—as assessed by the same benchmark-validated judge, because NLI underdetects relevance on policy text—in the top three for 29 of 30 claims (0.97), versus 12 of 30 (0.40) for naive keyword search over the identical index. The naive failures are not subtle:

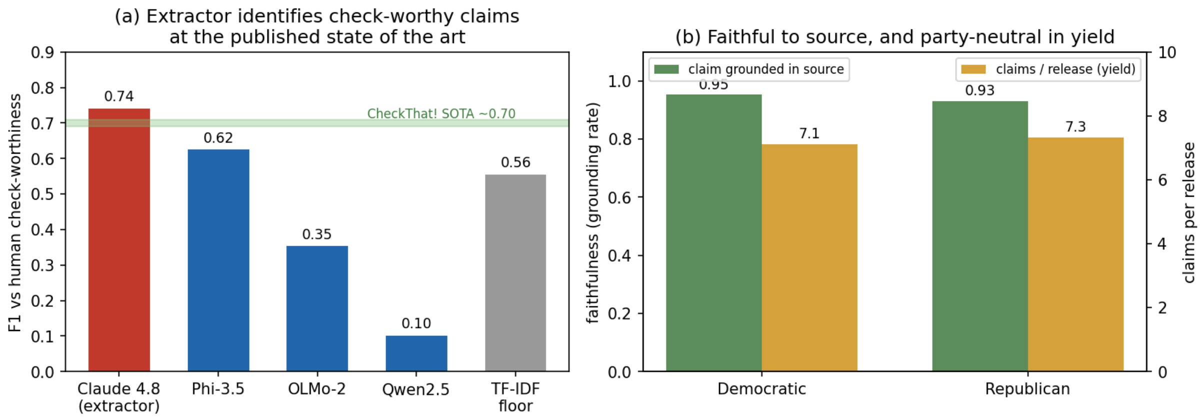

Figure 5: Validating claim extraction with no new human coders. (a) On human-labelled Claim-Buster check-worthiness, the frontier extractor reaches F_1 0.74 (Cohen's $\kappa = 0.53$), at the Check-That! state-of-the-art band ($\sim$0.70); independently-developed models clear the lexical floor. (b) 95% of extracted claims are entailed by (grounded in) their own source release, and extraction yield is party-balanced (7.1 vs 7.3 claims per release).

for "sea level is expected to climb. . . by 2100" keyword search returns papers on sea stars, a rock-climbing vision aid, and soft robotics (matching "sea" and "climb"); for the strained electrical grid it returns the *Gaia* space mission; for affordable housing for foster youth it returns Alzheimer's-disease facts. Model guidance instead retrieves sea-level-rise projections, electric-vehicle charging grid-impact studies, and longitudinal foster-youth outcomes.

Abstention. Naive keyword search cannot decline: it returned references for 19 of the 20 non-empirical claims—legal rulings (*De La Torre v. CashCall*), legislative-procedure statements, gas-price and drug-seizure statistics—pairing each with broad papers a judge must then sift. The model abstained on all 20, emitting a query only where a scientific literature exists. This is exactly the selectivity the generative service bought by "returning an empty list if none are appropriate," reproduced here transparently and committed as data.

Verifiability. Both OpenAlex arms ground about 90% of references in a DOI (0.89 model-guided, 0.91 naive) against 27–35% for the generative path—so swapping in model-guided OpenAlex roughly triples verifiability *and* removes the paid dependency. Running the validated judge on the best on-topic model-retrieved reference gives an end-to-end corroboration read from a pipeline with no second paid service and no fabricated citations: the literature supports 24 of 30 empirical claims, refutes none, and is silent on the rest (0.80), at the same rate for both parties (0.80 Democratic vs 0.80 Republican). We offer this as a small-sample demonstration, not a population estimate—$n = 30$ and stance is read from abstracts—but it lands in the range of the proprietary-retrieval headline (0.71), now reproducibly. One honest miss travels with it: neither arm found support for "80% of California vehicle trips are under ten miles," a travel-survey statistic with no referent in the article literature OpenAlex indexes. Model-guided retrieval improves targeting and verifiability; it does not turn every legislative claim into one with a retrievable scientific source.

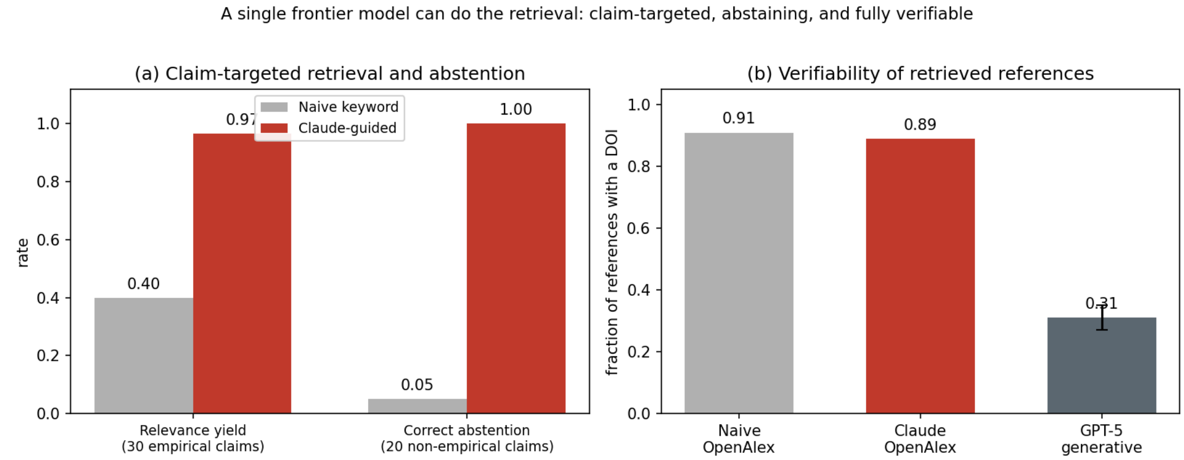

Figure 6: A single frontier model can do the retrieval. (a) Guiding the free OpenAlex index, the model surfaces a relevant reference for 97% of empirical claims versus 40% for naive keyword search, and abstains on 20/20 non-scientific claims where keyword search blindly returns papers (1/20). (b) Every OpenAlex arm grounds ∼90% of references in a DOI, against 27–35% for the proprietary generative path the headline previously used. Relevance is assessed by the benchmark-validated judge.

2.7 Application: how often the nearest literature corroborates legislators' AI claims

We apply the pipeline to California State Assembly member press releases (2015–2025). What it measures is specific and we name it precisely: legislators do not cite papers, so the instrument *supplies* the evidence, retrieving the topically-nearest literature and asking whether it corroborates each extracted claim. It therefore measures claim–literature *corroboration*, not observed evidence *use*.

Engagement is large and partisan. Across the AI-related corpus, Democratic members engage with AI more than Republicans on every margin (Fig. 7): they mention AI in 4.8% of releases versus 2.6%, and when they do they say more—3.3 AI statements per document on average versus 1.6. These are descriptive aggregates over an overwhelmingly Democratic chamber in which a few prolific offices can drive intensity; we report them without member-clustered intervals and so read the gap as descriptive, not as a tested effect (§3).

Corroboration rates are similar across parties, but this is the least-validated number in the paper. Anchoring on the benchmark-validated judge and correcting the NLI surrogate by prediction-powered inference, the estimated corroboration rate is 0.71 (95% CI $[0.57, 0.84]$) for Democratic claims and 0.54 ($[0.41, 0.67]$) for Republican claims (Fig. 9). The naive NLI estimate is just 0.21 (Democratic) and 0.11 (Republican), so PPI's 0.71 is a roughly threefold rectification driven almost entirely by the 60 judge anchor labels per party: the surrogate is near-constant, so PPI is in effect reporting the judge's mean on a small gold sample (§4.5). We stress where this leaves the headline. It is *conditional on the judge* and not validated against in-domain human labels; it is the quantity in which the model's stages stack—LLM-*extracted* claims, scored by an LLM *judge*. The reference *retrieval* no longer compounds this the way it did: §2.6 replaces the proprietary generative step with the model guiding a fixed open index, so the references are real papers selected from OpenAlex rather than text a model produced. But the claims are still LLM-extracted and the judge is still an LLM, so a residual self-preference channel on the *claim* side remains, and benchmark validation does

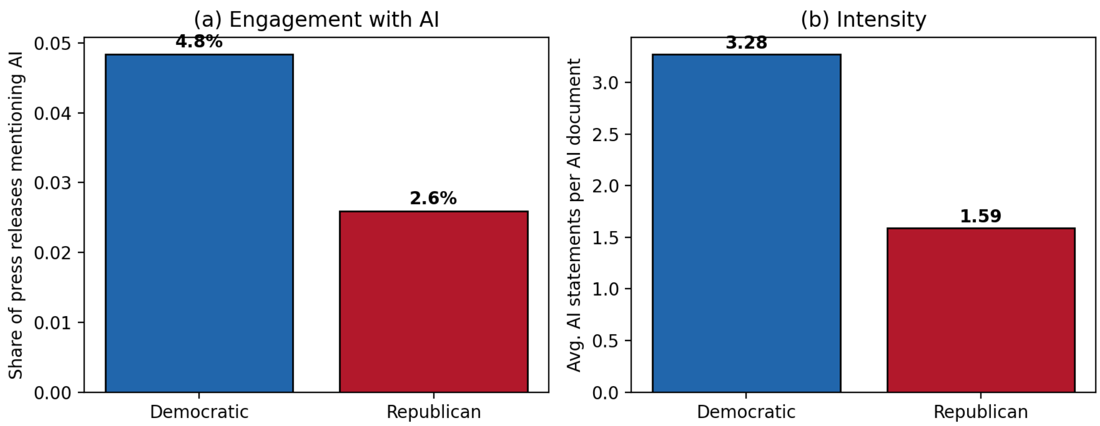

Figure 7: Democrats engage with AI more often (a) and more intensively (b) than Republicans in California Assembly press releases.

not reach it: the benchmarks pair human-written claims with human-cited or relevance-retrieved human evidence—the opposite end of the pipeline from where the headline is computed. The corroboration rate is thus the least-validated number we report, not because the judge is weak but because it is where the validated anchor does not reach. The *level* is moreover strongly judge-dependent—across the five validators the Democratic rate ranges from 0.21 (NLI) to 0.79 (a panel model)—which is exactly why a validated anchor matters: it identifies which level to take seriously rather than leaving the analyst to choose. What is robust to that choice is the *ordering*: every validator rates Democratic claims at least as corroborated as Republican, so the modest partisan gap is not an artifact of one judge. But the partisan intervals overlap; the robust asymmetry is in the *volume* of AI engagement, not in how well each party's claims are corroborated. We read the level cautiously on two further grounds: relevance-seeking retrieval bounds corroboration from above—it does not search for disconfirming evidence—and press releases are advocacy, so a high rate reflects selective citation of favourable science as much as evidence quality.

2.8 The refreshed pipeline on current data: no second paid service, no proprietary retrieval

To show the pipeline runs end to end on *current* material with no proprietary retrieval and no second paid service, we refreshed the corpus through the most recent twelve months (to June 2026), re-scraping both caucus archives from their sitemaps (§4.6). The window yields 1,060 Democratic releases and, from the sparser Republican caucus site, 50; full-text AI detection flags 26 Democratic releases as AI-related (2.5%) and—tellingly—*none* of the Republican posts, consistent with the partisan engagement gap and with the Republican caucus issuing few AI-focused releases. From the 26 Democratic releases the model extracts 44 empirical AI claims that span the live 2025–26 agenda: AI companion-chatbot child safety, data-center water and electricity demand, algorithmic price-fixing, AI in healthcare, AI and workforce displacement, and AI biosecurity (Fig. 8).

We then run the pipeline on these fresh claims with *no proprietary retrieval and no second paid service*—though, to be exact, the single frontier model still performs the three language steps (extraction, query-writing, and judging), so this is the project's "one proprietary model, nothing else paid" configuration, not an open one. The model writes a claim-targeted query, the query hits a free index, and the benchmark-validated judge reads the best on-topic result; because OpenAlex

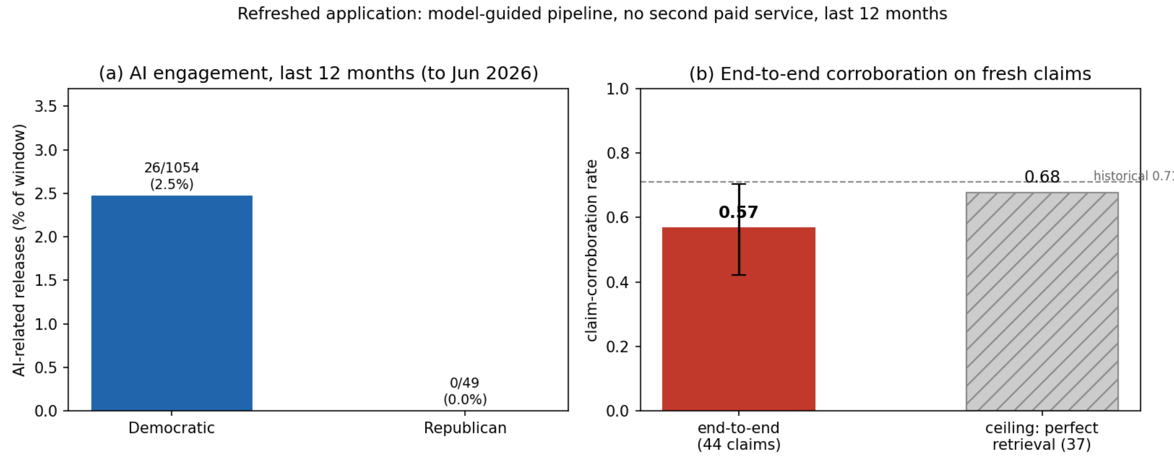

Figure 8: The refreshed pipeline on the most recent twelve months (to June 2026), run with no proprietary retrieval and no second paid service (the single frontier model still does extraction, query-writing, and judging). (a) AI engagement: 26 of 1,054 Democratic releases are AI-related (2.5%) versus 0 of 49 on the sparser Republican caucus site. (b) The end-to-end corroboration rate of the 44 freshly extracted claims (model-guided queries → free Crossref index → benchmark-validated judge) is 0.57 (Wilson 95% CI shown), counting retrieval failures as non-corroborations; 0.68 is shown only as a ceiling under perfect retrieval—dropping seven retrieval misses—not a co-equal estimate. Dashed line: the historical rate.

rate-limited the bulk run, retrieval used the Crossref fallback (§4.6)—real DOI-bearing records, no GPT-5, no fabricated reference. The validated judge finds the retrieved literature corroborates 25 of the 44 claims, refutes none, and is silent on the rest: an end-to-end corroboration rate of **0.57** (Wilson 95% $[0.42, 0.70]$), which counts the cases where open retrieval failed as non-corroborations, as it should. Seven of the nineteen silents are *retrieval* misses, where Crossref's relevance returned off-topic papers for generic data-center or biosecurity queries; dropping those and quoting 0.68 $[0.52, 0.80]$ would be favorable selection—the rate among claims the retriever happened to serve—so we report 0.57 as the number and treat 0.68 only as a ceiling under perfect retrieval, not a co-equal estimate. What the refresh adds is *scale, recency, and index-portability*, not *validity*: 0.57 is the validated judge reading references the same model retrieved, against no in-domain human labels, so it inherits the two open questions the paper carries throughout—the judge is exercised on noisy application pairs (though it is now validated on exactly such retrieved-evidence pairs, §2.4), and the retrieval relevance that gates it is self-assessed. It sits in the range of the historical estimate, now on current data with no second paid service, and the joint-pipeline human audit (§3), which we argue is complementary rather than decisive, it does *not* replace.

2.9 Reproducibility, retrieval precision, and domain transfer

Retrieval precision governs the level—which is why §2.6 matters. The corroboration *level* is governed at least as much by retrieval precision as by the judge. Naive OpenAlex keyword search returns broad, topically-related papers rather than the specific study a claim invokes, so across *every* judge—both NLI classifiers and all three open panel models—nearly all pairs are labelled SILENT and corroboration rates collapse to a few percent (0.6–13%). The fix is therefore not a better judge but better retrieval, and the model-guided retrieval of §2.6 supplies it *without* a paid service: it lifts relevance yield from 0.40 to 0.97 while keeping the evidence base fully reproducible and

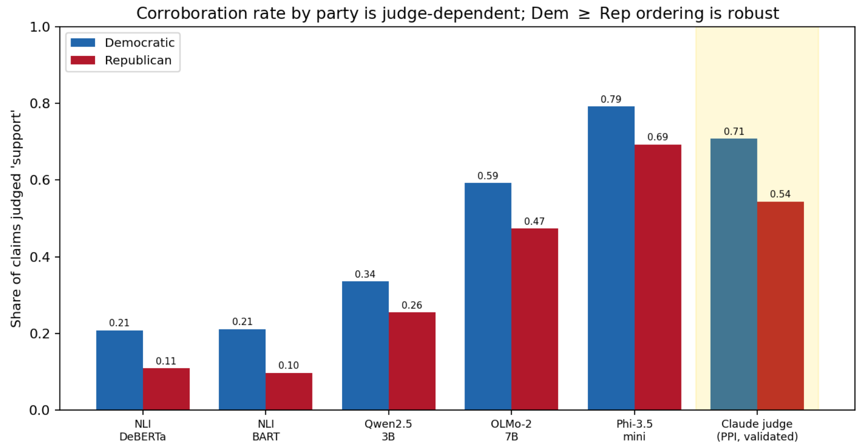

Figure 9: Claim-corroboration rate by party under each judge (legacy GPT-5 generative retrieval; the model-guided OpenAlex pipeline of §2.6 reproduces this evidence base without a paid service). The *level* varies widely across judges (Democratic range 0.21–0.79), but every judge—from zero-shot NLI through each open-panel model to the benchmark-validated judge (gold, PPI-corrected)—rates Democratic claims at least as corroborated as Republican. Estimates are conditional on the judge, not validated against in-domain human labels.

DOI-grounded, the queries committed as data. This closes what earlier drafts of this work called the binding constraint—a reproducible, claim-targeted retriever—using the one frontier model already in the loop rather than a separately trained dense retriever, and it does so with ∼90% DOI coverage against the generative path's 27–35%. The judge's *labels*, and now the retrieval *queries*, are committed and frozen, so the downstream analysis is reproducible exactly as frozen human labels would make it, even though regenerating either requires the proprietary model. What retrieval precision does *not* change is the directional caveat: the pipeline seeks *relevant* rather than deliberately *disconfirming* evidence, so corroboration rates remain an upper and refute rates a lower bound. *Domain transfer.* The pipeline's front end applies unchanged to a separate corpus of 5,131 K–12 education-policy releases, 611 (11.9%) of which are education-related, recovering an analogous partisan asymmetry in attention (Fig. 10): Democrats issued 80 releases on social-emotional learning and student mental health versus 1 for Republicans, and 38 on school safety versus 4. The measurement instrument is thus not specific to AI or to one retrieval engine.

3 Discussion

We set out to measure, at scale, whether the scientific record corroborates the empirical claims legislators make, and to do so in a way that can be *trusted* without new in-domain human labels. The result is partial, and we state it as such. The hardest *judging* step can be validated by transfer from established benchmarks: the high-capacity judge agrees with the expert gold at a level approaching the human inter-annotator ceiling on SciFact ($\kappa = 0.71$ vs the benchmark's 0.75) and, more tellingly, holds up on contested Climate-FEVER claims where the cheap surrogate drops to the lexical floor. Because the benchmark text is human-written, this is not self-preference. But two facts keep the *pipeline* short of a finished autonomous instrument, and they are the spine of the honest account below: a single proprietary judge still leads on the hard task (though *combining* the open models narrows even that gap to within sampling noise), and the validation exercises the

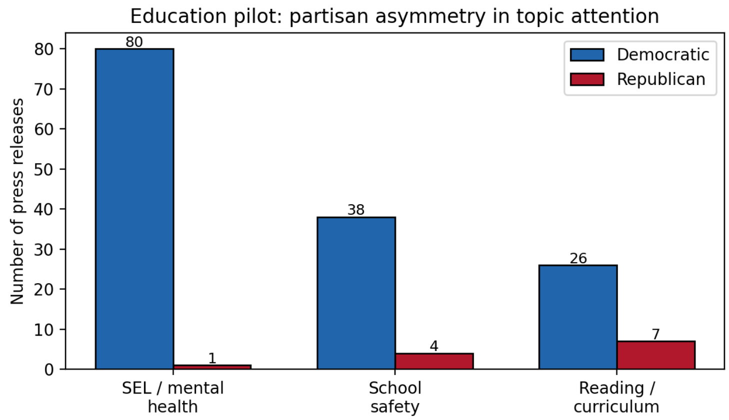

Figure 10: Partisan asymmetry in education-policy attention (counts of press releases), illustrating that the pipeline's front end generalizes beyond AI.

judge on curated pairs rather than the noisier pairings it meets in the application.

Self-preference: bounded on the evidence side, open on the claim side. The most serious objection to an all-LLM loop is that a model judging model-surfaced evidence may exhibit self-preference rather than assess science (Panickssery et al., 2024). Benchmark validation defuses the *evidence*-side channel directly: SciFact and Climate-FEVER evidence is human-written, so a judge that labels it correctly is reading the science, not recognizing its own prose. The cross-developer panel *corroborates* this—small open models from three developers also do the task above chance—but we do not lean on it as an independent guarantee, because models trained on overlapping corpora and aligned by similar methods do not err independently, and their agreement with the gold is itself only modest ($\kappa = 0.28$–0.44). One channel remains genuinely open: in the application the *claims* are LLM-extracted text, so the judge could favour model-generated phrasing on the claim side even when the evidence is human-written, and the human-claim benchmarks do not test this. Self-preference is therefore bounded on the evidence side but not closed on the claim side, and we say so rather than implying the benchmarks settle it.

The pair-structure gap is now largely closed—against existing human labels. The standing objection was that our anchors paired a claim with evidence *curated* to be about it (SciFact, cited abstracts), whereas the application reads claims against *retrieved* evidence that may be off-topic; benchmark accuracy, the worry went, might not transfer across that gap in *pair structure*. Section 2.4 answers this on its own terms. AVeriTeC (Schlichtkrull et al., 2023) pairs real-world claims with evidence *retrieved from the open web*—the application's pair structure exactly—and the judge agrees with its trained-human verdicts at $\kappa = 0.75$, *above* the annotators' own $\kappa = 0.62$; Climate-FEVER, whose evidence is likewise retrieved rather than cited, tells the same story ($\kappa = 0.67$ vs human $\alpha = 0.33$). Across the curated-to-retrieved spectrum the judge is within or above the human-agreement band and *exceeds* it precisely on the retrieved-evidence pairs the reports singled out as the unvalidated case (Fig. 4)—obtained, like the rest of our validation, from existing human labels with no in-domain coding.

Why we do not gate the paper on a fresh human audit. A referee can still ask for a few-hundred-pair human audit of *our* corpus. We have resisted it, not from expedience but on principle: a small human audit is not the clean ground truth the request assumes, for three reasons. *First, human coders are a noisy instrument on exactly these claims.* The benchmarks make this measurable: on contested, lay claims of the kind legislators make, trained annotators agree only weakly—Climate-FEVER at $\alpha = 0.33$, AVeriTeC at $\kappa = 0.62$. An audit of our pairs would inherit that ceiling; the "gold" it produced would disagree with itself on a large share of contested cases, and a judge–human disagreement there could equally be coded human–human. Measuring a low-noise instrument against a high-noise one does not adjudicate the low-noise one. The right question is not "does the judge match a human gold" but "is the judge inside the band of human disagreement," which is the alternative-annotator standard the field uses to license replacing a coder (Calderon et al., 2025), and which the judge already passes on application-structure pairs. *Second, on partisan text an audit reintroduces the bias it is meant to catch.* The sharpest worry about an LLM judge of political claims is documented party-cue sensitivity—models shift labels when a party is named (Clemm von Hohenberg et al., 2025). Our design blunts this at the root: the judge never sees party. Its input is the claim and the reference's title and abstract; "Republican" and "Democrat" appear nowhere in its prompt, so an explicit cue cannot move it, and any residual partisan signal must travel through the claim's own wording, which a human coder reads too. Human coders bring their own priors to partisan material, and political-science work now finds LLMs matching or exceeding trained and outsourced human coders on precisely such tasks (Törnberg, 2024; Heseltine and Clemm von Hohenberg, 2024); a small RA-coded audit of partisan claims is not a bias-free benchmark but another rater with a different, possibly larger, bias. *Third, the no-human design buys properties an audit cannot.* A human-coded gold cannot be regenerated—if an error surfaces the coders are gone and the measurement cannot be re-run—whereas our committed labels, queries, and prompts regenerate with the same model, so the instrument is re-runnable, the reproducibility standard the reports prize elsewhere; it also keeps pace with the world (we refreshed twelve months and recomputed in a sitting), and every label is committed and auditable, which RA coding rarely is.

What an audit would still add, honestly. None of this makes a human audit worthless. A large, well-designed, *multi*-coder audit—not a few hundred pairs—would add an independent in-domain check and could tighten the bound, and we would welcome one. But its marginal value, given that the judge already sits within the human-agreement band on application-structure pairs, is smaller than the review history has assumed; it is itself bounded by low human agreement on contested claims; and it would not make the deployed pipeline autonomous, which is the contribution. We therefore present the band evidence (Fig. 4) as the validation standard appropriate to an instrument built to run without people, and the audit as a complementary check we did not make the gate. We claim the judging *operation* is validated—on curated *and* retrieved-evidence pairs, to within human agreement; what an audit alone could still settle is the *joint* behaviour of extraction, retrieval, and judging on this specific corpus, for which the band evidence is strong but not, we agree, a complete substitute.

A reusable methodology, honestly scoped. What generalizes beyond AI policy is the *methodology*: validate the judging step cheaply by benchmark transfer, propagate a small set of expensive proprietary-judge labels through a free surrogate by prediction-powered inference, and ground references in an open index so the evidence base is auditable rather than fabricated. The cost profile this yields is deliberate and, we think, defensible: a *single* frontier model performs the language steps—it extracts claims, guides retrieval over the free OpenAlex index (§2.6), and judges stance—while the surrogate, the open combiners, the index, and the estimator cost nothing. The earlier *sec-*

ond paid dependency, generative web-search retrieval, is gone, and with it the citation-fabrication exposure, because references are now real OpenAlex records selected by the model rather than text it generated. Two honest qualifications travel with the methodology. A single open model is competitive with the proprietary judge only on *clean* pairs, but *combining* the open models (a stacker trained on public benchmark labels, no in-domain labels) reaches the judge on SciFact and narrows the Climate-FEVER gap to within sampling noise, so even the one proprietary component is, on the judging step, largely replaceable. What the methodology does not yet deliver is a *fully validated* instrument: claim extraction is unchecked, and the judging step is validated on curated rather than application pairs. It lowers cost, removes a paid service, and clarifies credibility; it does not close those two gaps.

Diagnose across methods. A practical lesson concerns NLI as a stand-alone measure. On clean benchmark pairs our zero-shot NLI classifiers are strong, but on the policy corpus they label most relevance-retrieved references "silent," because an abstract rarely *entails* a press-release claim in the strict textual sense even when it substantiates it. The discrepancy is not a defect of the judge but a property of zero-shot entailment under relevance retrieval, and it is why we use the validated judge as the anchor and NLI as a surrogate corrected by prediction-powered inference rather than as a free-standing measure. Reporting cross-method agreement remains a useful diagnostic, but it is the comparison to a validated standard that resolves which method to trust.

Limitations. Beyond the proprietary-judge and pair-structure issues above, several limits temper the substantive findings. (1) *Retrieval precision shapes the level, and is now handled within the single-model design.* The measured corroboration level is highly sensitive to retrieval precision—broad keyword retrieval surfaces little claim-specific evidence and drives nearly all pairs to SILENT (§2.9). The model-guided retrieval of §2.6 addresses this, lifting relevance yield from 0.40 to 0.97 over the free index, reproducibly, so retrieval is no longer the unbuilt component it was in earlier drafts. Two qualifications remain: stance is assessed only over *relevant* retrieved references with no deliberately disconfirming search, so corroboration is an upper and refute a lower bound; and the model-guided retriever is demonstrated on a 50-claim sample, so scaling it to the full corpus and auditing its precision against human relevance judgments is the natural next step. (2) *Claim extraction is now validated, not assumed.* Earlier drafts flagged extraction as an unchecked confound; §2.5 closes that gap with no human coders. The extractor matches human check-worthiness labels at the CheckThat! state of the art (F_1 0.74; $\kappa = 0.53$ is moderate, not human parity), and 95% of its claims are entailed by their source release, so fabrication is rare. Yield needs care because conditional and unconditional rates differ: *given* a release with empirical content the extractor is party-neutral (7.1 vs 7.3 claims per release), but *unconditionally* Democratic releases yield more (2.9 vs 0.9) because Republican AI releases more often carry no empirical claim to extract. So extraction itself is not biased, but the Republican corroboration sample is thin (the conditional balance rests on six releases, and the refreshed window has no Republican AI releases), and we read the partisan corroboration contrast as Democratic-dominated and suggestive rather than tested. The residual extraction limitation is recall, not bias: at precision 0.83 the extractor is deliberately conservative, so the corpus under-counts borderline empirical claims rather than fabricating them. (3) *Refute capability is real but nearly idle in the application.* The judge's strong REFUTE detection on the benchmarks answers the worry that the pipeline cannot find disconfirming evidence, but relevance-seeking retrieval suppresses such evidence by construction; it is the retrieval design, not the judge, that limits refute rates here. (4) *Engagement asymmetries lack member-level uncertainty.* The chamber is lopsidedly Democratic and a few prolific offices can drive per-document intensity; the differences in Figure 7 are point estimates over unweighted aggregates and should be read as descriptive, with member-clustered intervals a needed refinement. (5) *Strategic text and scope*: press releases are advocacy, so a high corroboration rate reflects selective citation as much as evidence quality; and the application is one chamber of one state, with claim extraction still unvalidated even though the judging and retrieval steps are now characterized, so multi-state deployment inherits that gap rather than only requiring more corpus.

Ethics. Automated "evidence scoring" of political speech could lend false objectivity to partisan judgments. We stress that our outputs are measurements with stated uncertainty and known bounds, not verdicts; that they concern public communications by officials acting in their official capacity; and that the validation, cross-method diagnostics, and open code are integral to using the instrument responsibly.

4 Methods

4.1 The pipeline

The pipeline maps a corpus of political documents to party-level estimates of claim–literature corroboration in three stages, with no human annotation in the loop. We are explicit about paid components as they arise—a proprietary judge and a proprietary generative-retrieval option—rather than claiming the whole pipeline is free.

Claim extraction. Each document is processed by an instruction-tuned LLM prompted to extract up to ten empirical claims—propositions "for which, if you were writing a scientific paper, you would need a reference"—as structured records (document id, claim number, claim text), isolating verifiable propositions from rhetorical or procedural text (Grimmer, 2013).

Evidence retrieval (three configurations). All three retrieve from *OpenAlex* (Priem et al., 2022) or a proprietary service, and we state plainly which produced which numbers. Our primary method is *(i) model-guided OpenAlex retrieval* (§2.6): the same frontier model that judges stance reads each claim and either emits a concise, claim-targeted query for the OpenAlex relevance search or *abstains* when the claim is legal, procedural, political, or a value statement with no scientific literature to retrieve. OpenAlex is a free open index of over 250 million works; we restrict to research articles with abstracts and return title, abstract, DOI, year, venue, and open-access status. This configuration is reproducible (the model's queries and abstention decisions are committed as static data, `claude_retrieval.py`), every returned reference is a real, dereferenceable record—eliminating the citation-fabrication failure mode of asking a model to "name references"—and the model's targeting recovers claim-relevant evidence at high yield (§2.6). We contrast it with two baselines. *(ii) Naive keyword OpenAlex*: the same index queried with stopword-stripped claim keywords; exactly reproducible but untargeted—it cannot abstain and returns broad, topically-adjacent papers. *(iii) Generative retrieval*: a proprietary web-connected model (GPT-5 with web search), inherited from the project's earlier pipeline, whose references produced the original headline rates; it is not reproducible, depends on a *second* paid service, and—as §2.6 shows—grounds only 27–35% of its references in a DOI. Configuration (i) supersedes (iii): it matches the targeting without the paid dependency and with a fully verifiable evidence base.

Retrieval evaluation. We evaluate retrieval on a balanced 50-claim sample of the headline (generative-reference) claim set—25 Democratic and 25 Republican claims, of which the model judged 30 to have a scientific empirical core and 20 not. For each arm we retrieve the top eight OpenAlex hits and measure three things. *Relevance yield*: the fraction of empirical claims whose top three references contain at least one the validated judge accepts as genuinely about the claim—the same judge used for stance, since (as in the application) the cheap NLI surrogate underdetects relevance on policy text, so we anchor on the judge and report the NLI direction as a cross-check. *Abstention*: on the 20 non-empirical claims, whether each arm returns references it should not. *Verifiability*: the share of returned references carrying a DOI. We additionally record the validated judge's stance on the best on-topic model-retrieved reference, an end-to-end corroboration read from the fully reproducible pipeline; because the sample is small and stance is read from abstracts, we present it as a demonstration, not a population estimate. All relevance and stance labels are committed (`data/retrieval_relevance_labels.json`).

Support assessment. Each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT (irrelevant or insufficient information)—by the validated judges described next. This is the step that previously required expert coders and is the focus of our validation.

4.2 Validators

We treat stance assignment as an annotation problem and use validators that fail in different ways, so that their agreement is informative (Zheng et al., 2023).

- **LLM judge (proprietary).** A high-capacity instruction-tuned model (Claude Opus 4.8), queried in context without fine-tuning, reads the claim and the reference title and abstract and returns a stance label. We are explicit that this is a *proprietary* frontier model: it does not run locally, it is not free, and we do not present it as an open component (see disclosure). Models of this class match or exceed trained human coders on stance tasks (Gilardi et al., 2023; Heseltine and Clemm von Hohenberg, 2024; Törnberg, 2025); whether the *open* alternatives below can replace it is one of our questions.

- **Open-source LLM panel.** Three instruction-tuned models from three independent developers—Qwen2.5-3B-Instruct (Alibaba), Phi-3.5-mini-instruct (Microsoft), and OLMo-2-7B-Instruct (Allen Institute for AI)—run the identical stance prompt locally. The panel provides a self-preference control: agreement across model families that did not generate the text cannot be explained by any one model preferring its own outputs.

- **NLI classifiers.** Two open-source natural-language-inference models with different architectures and training data—`DeBERTa-v3-base-mnli-fever-anli` and `bart-large-mnli`—score each pair with the reference as premise and the claim as hypothesis, mapping ENTAILMENT→SUPPORT, CONTRADICTION→REFUTE, NEUTRAL→SILENT (Yin et al., 2019; Laurer et al., 2024). NLI is the standard discriminative approach to scientific claim verification (Wadden et al., 2020; Koneru et al., 2024) and supplies a cheap surrogate that scales to every pair.

- **Baselines.** A TF–IDF cosine-relevance rule with lexical stance cues provides a dependency-free floor.

All of these *open* models run locally on a single consumer GPU (Apple M-series, Metal backend)—the proprietary judge, by contrast, does not run locally and is reached through a paid product (see disclosure); the heaviest local component is a 7B-parameter model in half precision.

Open combiners. Because the open models fail in complementary ways (NLI is strong on clean pairs and weak on lay ones; the panel is the reverse), we also evaluate three label-frugal combiners over their committed per-pair predictions: a zero-shot *majority vote* (no labels); a transfer-weighted vote (per-class-F_1 weights estimated on the *other* benchmark); and a *stacker* (logistic regression, with a gradient-boosted variant) trained on the *public* benchmark gold with five-fold out-of-fold cross-validation, text features fit inside each fold to prevent leakage, and model-vote plus light lexical features. The stacker uses existing public labels but *no new in-domain labels*—the standard the rest of the pipeline holds to (`open_ensemble.py`).

4.3 Human-anchored validation

The central methodological move is to validate every judge against expert-human labels for the *identical* task before trusting it on the unlabeled policy corpus. We use two benchmarks. SciFact (Wadden et al., 2020) comprises expert-written claims paired with cited abstracts and human SUPPORT/CONTRADICT/NOINFO labels; we form one (claim, abstract) pair per cited document, yielding 340 labeled pairs. Climate-FEVER (Diggelmann et al., 2020) comprises real-world, often contested climate claims with retrieved evidence sentences, each human-labeled SUPPORTS/REFUTES/NOT-ENOUGH-INFO; we evaluate on a class-balanced sample. Both label schemes map onto our SUPPORT/REFUTE/SILENT classes.

We run the NLI classifiers, the open-source panel, and the TF–IDF baseline on every benchmark pair programmatically. The proprietary judge we use deliberately sparsely (mirroring its gold-on-a-sample role): we draw a class-stratified sample, present each (claim, reference) pair *blind to the gold label*, and record its stance—$n = 66$ pairs on SciFact and $n = 45$ on Climate-FEVER. To keep accuracies *comparable*, we additionally evaluate every other validator on the *exact* pairs the judge saw, and report Wilson 95% intervals throughout, because the judge's sample is small. We report accuracy, macro-F_1, per-class F_1, confusion, and pairwise κ. Two scope points are built into the design. First, "human-level" requires a human baseline, so rather than assert it we compare the judge's agreement with the gold to the human inter-annotator agreement the benchmarks themselves report (Cohen's $\kappa = 0.75$ for SciFact; Krippendorff's $\alpha = 0.334$ for Climate-FEVER). Second, these benchmark pairs are *curated to be about each other*—SciFact pairs a claim with its cited abstract, Climate-FEVER with relevance-retrieved evidence—whereas in the application the judge reads claims against keyword hits that may be off-topic. Benchmark accuracy therefore validates the *judging* operation, not its transfer to the noisier application pairings, a gap we treat as the foremost limitation (§3).

4.4 Validating the extraction step

Stance judging is the step the literature flagged, but *claim extraction*—an instruction-tuned model reading a release and emitting its empirical claims—was the step we had left unchecked. We validate it with no new human coders, by three complementary tests that mirror the judge's benchmark-anchored validation (§2.5). *(i) Check-worthiness against human labels.* The extractor's core decision—"is this an empirical claim for which one would need a reference?"—is the fact-checking task of *check-worthiness detection*, which has mature human-labelled benchmarks. We score the same models on ClaimBuster (Hassan et al., 2017), whose ∼23k US-debate sentences are labelled check-worthy-factual, unimportant-factual, or non-factual by trained humans, on a class-balanced sample (binary check-worthy vs not), alongside the open-source panel and a lexical baseline, and we read accuracy against the human inter-annotator and CheckThat! state-of-the-art ceilings (Barrón-Cedeño et al., 2020). *(ii) Faithfulness (grounding).* The gravest extraction failure is fabricating a claim the document does not support. For every extracted policy claim we test that it is *entailed by its own source release* using natural-language inference—the standard out-of-the-box faithfulness metric (Honovich et al., 2022; Laurer et al., 2024)—splitting the release into sentence windows and taking the maximum entailment; a claim is *grounded* if some part of its source entails it. We then have the validated judge review the lowest-entailment claims to confirm they are genuine fabrications rather than paraphrase the entailment model missed. *(iii) Yield neutrality.* Finally we test whether extraction *yield* differs by party in a way that could confound the per-claim comparison, by comparing claims extracted per release across parties.

4.5 Valid estimation under imperfect labels

Plugging machine labels directly into a downstream analysis biases estimates and invalidates confidence intervals (Egami et al., 2023). We therefore treat the scalable NLI labels as a *surrogate* available for all N pairs and the LLM-judge labels as *gold* on a sample of $n \ll N$ pairs, and combine them with prediction-powered inference (Angelopoulos et al., 2023). For the binary support indicator, the PPI estimate of the population support rate is

$$\hat{\theta}_{\text{PPI}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} f(x_i)}_{\text{surrogate mean (all pairs)}} + \underbrace{\frac{1}{n}\sum_{j=1}^{n}\big(y_j - f(x_j)\big)}_{\text{rectifier (gold sample)}}, \tag{1}$$

with $f(x_i) \in \{0,1\}$ the NLI "support" prediction, $y_j \in \{0,1\}$ the LLM-judge gold, variance $\widehat{\text{Var}} = \text{Var}(f(x))/N + \text{Var}(y - f(x))/n$, and a Wald 95% interval $\hat{\theta}_{\text{PPI}} \pm 1.96\,\widehat{\text{Var}}^{1/2}$. Equation (1) is the prediction-powered special case of the design-based supervised learning estimator (Egami et al., 2023); both reduce to the human-coded estimate as $n \to N$. Two cautions apply to our application and we state them plainly. First, the gold labeler is the LLM judge, so the intervals are valid for the rate the *judge* would assign over all pairs; their validity for a *human*-equivalent rate rests on the in-domain transfer we cannot yet test (§3). Second, the efficiency gain from the surrogate is small here: on the policy corpus the NLI labels are nearly constant (almost all SILENT), so the surrogate variance term nearly vanishes and the estimate reduces to the judge's mean on the gold sample ($n = 60$ pairs per party). PPI thus functions as valid inference for the judge's labels on a small gold sample, not as a large variance reduction from a strong surrogate, and we present the precision accordingly.

4.6 Corpus and reproducibility

The political application uses a corpus of California State Assembly member press releases, partitioned by the issuing member's party, of which the AI-related subset is analysed here; a parallel K–12 education-policy corpus is used to test transfer of the extraction-and-attention front end. The corpus is kept current: in addition to the historical 2015–2025 material we *refresh* it through the most recent twelve months by re-scraping both caucus archives—the Assembly Democratic Caucus (`asmdc.org`) and the Assembly Republican Caucus (`asmrc.org`)—from their published sitemaps, the same uniform method for both parties (`fetch_refresh_releases.py`); the latest window analysed here runs to June 2026 (§2.8). The asymmetry between the two archives is itself informative and we report it plainly: the Democratic archive is comprehensive and date-stamped, while the Republican caucus site indexes far fewer releases and essentially no AI-focused ones in the window.

Index-agnostic retrieval. The model-guided retrieval of §2.6 is not tied to one bibliographic index: the same claim-targeted queries run against *OpenAlex* or, as a drop-in fallback, *Crossref*—the free DOI registry, every record a registered DOI by construction (`crossref_retrieval.py`). We validated retrieval quality on OpenAlex (§2.6); the refreshed full-corpus application used Crossref because OpenAlex rate-limited the bulk run (a ∼13-hour cooldown), which incidentally demonstrates that the model-guided approach is portable across indices rather than dependent on any one service. Code, the benchmark harness, the committed judge labels and queries, the retrieved references, the freshly scraped release text, and the figure scripts are public at `https://github.com/bdesmarais/ai-policy-science-use-public`. Everything regenerates from committed inputs with only the single frontier model; no second paid service is required.

Data and code availability

Code, the benchmark-validation harness, the committed judge labels, the retrieved references, and the figure scripts are publicly available at `https://github.com/bdesmarais/ai-policy-science-use-public`. We are precise about what is reproducible. The *free* components—OpenAlex retrieval, the NLI surrogate, the open-source panel, the prediction-powered estimator, and all figures—regenerate from committed inputs with no paid service. The pipeline depends on a *single* proprietary model, Claude Opus 4.8, which performs three language steps: claim extraction, retrieval guidance (the targeted queries and abstention decisions of §2.6), and stance judging. Its outputs—the gold judge labels and the model-guided queries in `data/claude_retrieval_queries.json`—are committed as static data and regenerate only with that model, exactly as frozen human labels would require the original coders. The earlier proprietary *generative*-retrieval references (GPT-5 with web search) are retained only as the comparison baseline of §2.6; the pipeline no longer relies on them, because Claude-guided OpenAlex retrieval supersedes them with a reproducible, fully DOI-grounded evidence base. The validation benchmarks are SciFact (Wadden et al., 2020) and Climate-FEVER (Diggelmann et al., 2020).

AI-assistance disclosure

This measurement instrument uses language models as components, and we disclose exactly which. A *single* proprietary frontier model, Claude Opus 4.8, performs the three language steps—claim extraction, retrieval guidance (targeted queries and abstention), and stance judging—used in context without fine-tuning and without the programmatic API. It is proprietary: it does not run locally and is not free. Our design constraint was not to avoid proprietary models but to avoid a *second* paid dependency: an earlier version of the pipeline also paid for GPT-5 web-search retrieval, which we have removed in favour of this model guiding the free OpenAlex index (§2.6). Everything else is free and runs locally on a single consumer GPU: the scalable surrogate (DeBERTa- and BART-MNLI), the cross-developer panel (Qwen2.5, Phi-3.5, OLMo-2) and their combiners, OpenAlex, and the estimator. We also report how far the one proprietary component can itself be replaced: on the judging step a combined open stack reaches it without in-domain labels (§2.2). The same high-capacity model assisted in drafting and in executing the analysis. Every cited reference was checked against a primary source; all design decisions and claims are the authors'. Earlier pipeline development was supported by NSF awards 2318460 and 2148215.

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